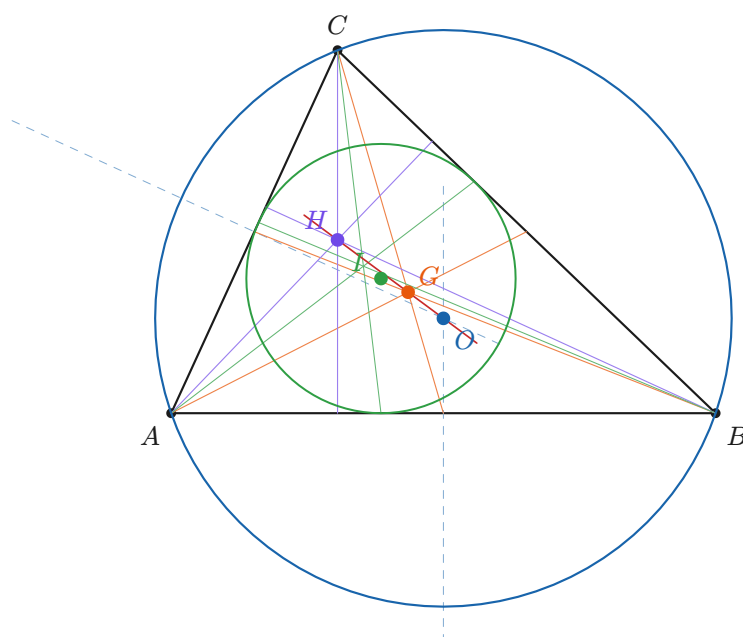


geomtools

✍️ FERGOUS Abdelhak

Drawing instruments for a geometry figure

ruler · pencil · set square · protractor · compass



Triangle ABC used throughout, its four centres, and Euler's line (OGH).

O circumcentre

perp. bisectors, compass

I incentre

angle bisectors, compass

G centroid

medians, ruler

H orthocentre

altitudes, set square

The Typst package `geomtools` lays the **geometer's instruments** on a figure: ruler, pencil, set square, protractor, compass — crisp like the LaTeX original, or “hand-drawn”. This guide introduces them, then uses them to **construct** the four centres of a triangle, with their markings (equal-length ticks, right angles, bisector arcs).

It is a port of Cédric Pierquet's `OutilsGeomTikZ`. No CeTZ: the tools are plain lists of polygons, arcs and labels.

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1 The package in two minutes

1.1 Local install

The package is not (yet) on Universe. Import it by path:

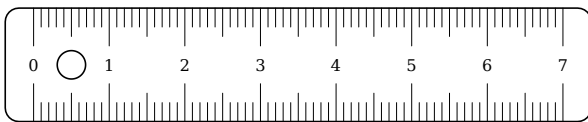
```
#import "path/to/geomtools/lib.typ": *
```

Everything the package exports lives in `geom`: you pass it a **list of primitives** — what each instrument **returns**, not Typst content.

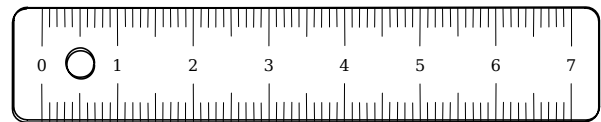
```
#geom({
  ruler(length: 10)
  pencil(at: (3, 2), rotate: -20deg)
})
```

Inside a `{ ... }` block, Typst **joins** arrays. You can also write `ruler(...) + pencil(...)` with `+`, which is more explicit, and that is what every listing in this guide does.

1.2 Two modes, one geometry



mode: "clean" — the default



mode: "rough"

```
#geom(ruler(length: 7))
#geom(ruler(length: 7), mode: "rough", roughness: 2, seed: 4)
#geom-rough(ruler(length: 7))
```

The wobble is **deterministic**: same seed, same figure on every compile. Numerals are never wobbled — a hand-drawn “3” would be a different typeface, not a shakier 3.

1.3 What you compose

| Function | Role on a construction | |—|—| | compass(from, to) | needle at from, lead at to — the opening is exact | | set-square | 30-60-90 set square, right angle at the origin | | ruler | zero at the origin, running right | | right-angle | right-angle **mark** (open square, not the tool) | | pencil | pencil, tip at the origin pointing up | | p-line p-arc p-circle p-label | the figure itself |

Common arguments: `at`, `rotate`, `scale`, `colour`, `fill`. The French keys of the original (*Longueur*, *Origine*...) map to English (`length`, `at`...).

Textbook convention

The solid stroke is the figure; the dash is the trace of the instrument — the one you rub out afterwards. Every primitive takes `dash: "dashed"`. A construction arc is therefore:

```
p-arc(A, 4.5, 20deg, 160deg, stroke: luma(120), dash: "dashed")
```

2 The canvas and the toolbox

Coordinates are in **centimetres**, y pointing **up** (mathematical orientation). The renderer flips the axis once.

2.1 Helpers already exported

`vadd` `vsub` `vmul` `vnorm` `dist` `vangle` `arc-pts` `circle-pts` `rect-pts` — plus the constructors `p-poly` `p-line` `p-circle` `p-arc` `p-label`.

`vangle((x, y))` is the bearing of the vector, 0° on $+x$, 90° on $+y$. That is the angle you pass to rotate.

2.2 Helpers in this guide

`helpers.typ` adds what a construction needs:

- `midp(A, B)`, `unit(v)`, `lerp(A, B, t)`, `foot(P, A, B)`
- `circ-inter(P, r1, Q, r2)` — the two intersections of two circles
- `line-inter(P1, P2, Q1, Q2)` — intersection of two lines
- `circumcenter` `incenter` `centroid` `orthocenter`
- `ticks(P, Q, n: 1)` — equal-length marks at the midpoint of $[PQ]$
- `right-angle` (from the package) and `ang-mark` — angle markings
- `square-on(at, along, toward)` — seats the set square on a line, right angle turned toward a point

The working triangle, used everywhere:

$$A = (0, 0), \quad B = (7.2, 0), \quad C = (2.2, 4.8)$$

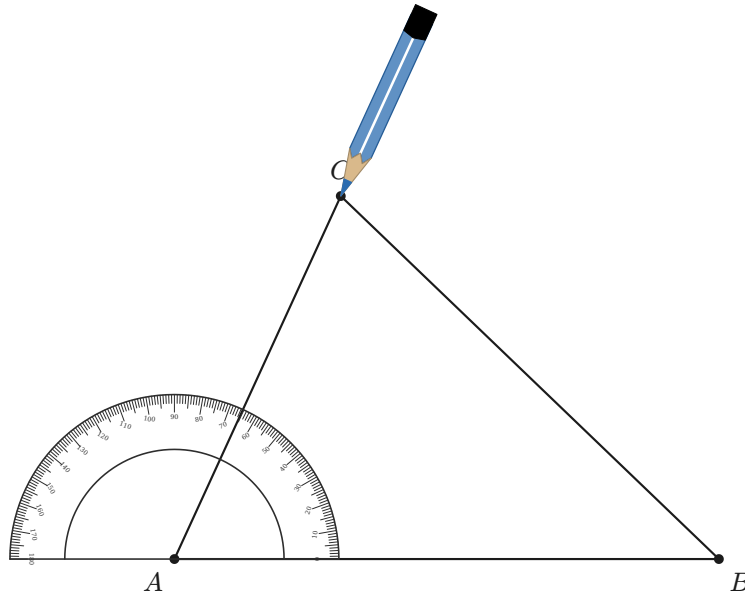
It is **scalene** and **acute**: the four centres are distinct and **interior**, and the feet of the altitudes fall on the sides (not on their extensions). AB lies on the x -axis, which makes the reading easier without making the figure special.

```
#let A = (0.0, 0.0)
#let B = (7.2, 0.0)
#let C = (2.2, 4.8)
```

3 Protractor, and a pencil at the end of the stroke

To **measure** $\angle BAC$, centre the protractor at A and align its base with (AB) . Side $[AC]$ meets the scale at the size of the angle (here $\approx 65^\circ$).

When you **draw a line**, it often helps to put a pencil at its end: the lead sits on the tip, the barrel follows the stroke. `pencil-tip(from, to)` does that — pencil's tip is at the origin and points up, hence `rotate: vangle(to - from) - 90deg`.



Protractor at A, base on (AB) . The pencil sits at the end of $[AC]$.

```
#protractor(at: A, rotate: 0deg, scale: 0.58, radians: false)
#pencil-tip(A, C, colour: rgb("#2B6CB0"), lead: rgb("#2B6CB0"))
```

`scale` shrinks the protractor (original radius 3.75 cm) without moving the centre. `radians: false` hides the $\frac{\pi}{6}$, $\frac{\pi}{4}$... band if you only want degrees. For a 0° – 360° disc: `full: true`.

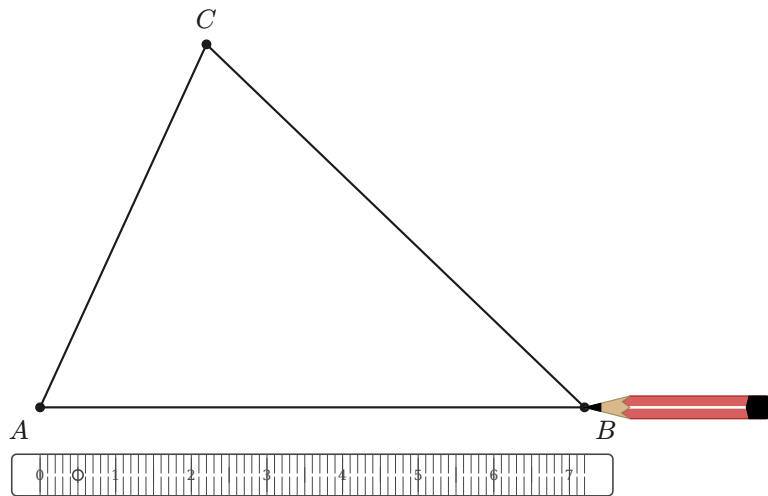
4 Circumcircle — with the compass

The **circumcircle** is the unique circle through A , B and C . Its centre O is the intersection of the **perpendicular bisectors**. A perpendicular bisector is built with the compass: two arcs of **equal radius**, larger than half the side.

What you mark

- equal-length ticks $AM = MB$ (the midpoint);
- **right angles** at the midpoint: the bisector is perpendicular to the side;
- at the end, $OA = OB = OC$ (the radius), optionally with three small arcs, though the circle itself is usually enough.

4.1 Step 1 — The triangle



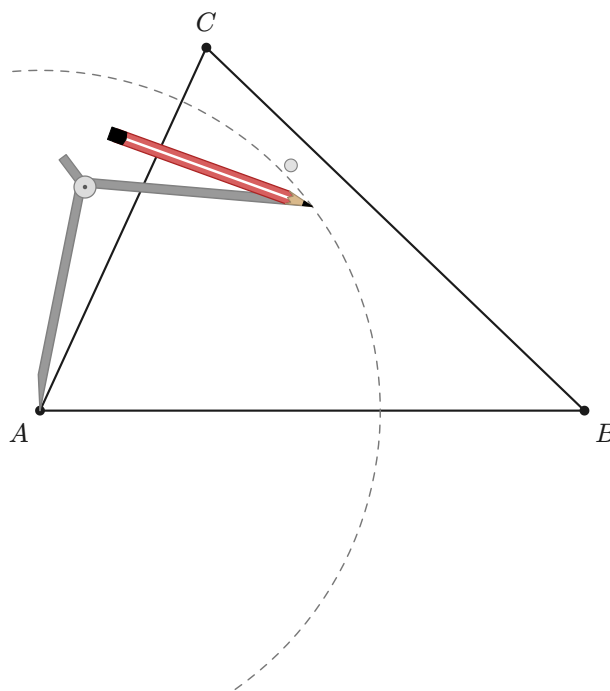
Draw AB with the ruler; the pencil sits at the end, at B.

```
#geom(
  p-line(A, B, stroke: black, weight: 1.35, role: "edge")
+ p-line(B, C, stroke: black, weight: 1.35, role: "edge")
+ p-line(C, A, stroke: black, weight: 1.35, role: "edge")
+ ruler(at: (0, -0.62), length: 7.2, width: 0.55, clamp: false)
)
```

clamp: false is essential: by default a ruler's width is silently floored at 1.5 cm, and a ruler you thought was thin sits 8 mm off.

4.2 Step 2 — Compass at A, opening larger than half the side

$AB = 7.2$, so $AB/2 = 3.6$. Take 4.5 cm. The arc must be long enough to meet its twin from B, **on both sides** of (AB) .



compass(A, P) — needle at A, lead through a point of the arc.

```
#let rAB = 4.5
#p-arc(A, rAB, -55deg, 95deg, stroke: luma(120), dash: "dashed")
#compass(A, (4.5 * calc.cos(40deg), 4.5 * calc.sin(40deg)), scale: 0.58)
```

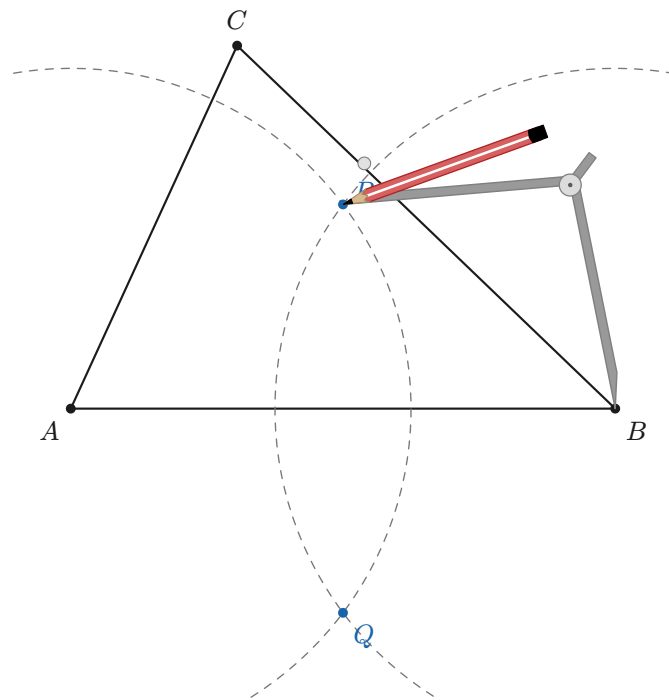
The compass **really opens** on the two points:

$$\text{half-angle} = \arcsin((|to - from|)/(2 \text{ leg scale}))$$

scale shrinks the instrument **without** moving the feet — essential if it is to fit the figure. flip: true swings it to the other side of the segment when it would cover the triangle.

4.3 Step 3 — Same opening, needle at B

The two arcs meet at P and Q. **Do not change the opening**: that is the whole marking $AP = BP = AQ = BQ$.



Second arc, **same radius**. P and Q are equidistant from A and from B .

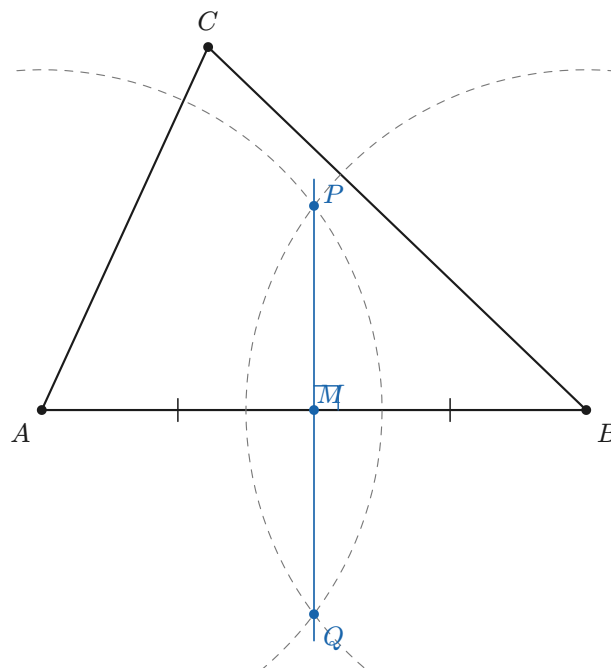
Intersections are computed, not guessed:

```
#let (P, Q) = circ-inter(A, rAB, B, rAB)
#compass(B, P, scale: 0.58, flip: true)
```

`circ-inter` (in `helpers.typ`) is the classical intersection of two circles: project onto the line of centres, then step off $h = \sqrt{r_1^2 - a^2}$.

4.4 Step 4 — The perpendicular bisector of $[AB]$

The line (PQ) is the perpendicular bisector: it cuts $[AB]$ at its midpoint M and is perpendicular to it. Mark both facts.



Marking: $AM = MB$ (one tick) and $\angle PMB = 90^\circ$ (open square).


```
#let M = midp(A, B)
#line-ext(P, Q, stroke: blue, dash: none)
#ticks(A, M, n: 1) + ticks(M, B, n: 1)
#right-angle(at: M, rotate: 0deg, size: 0.32, colour: blue)
```

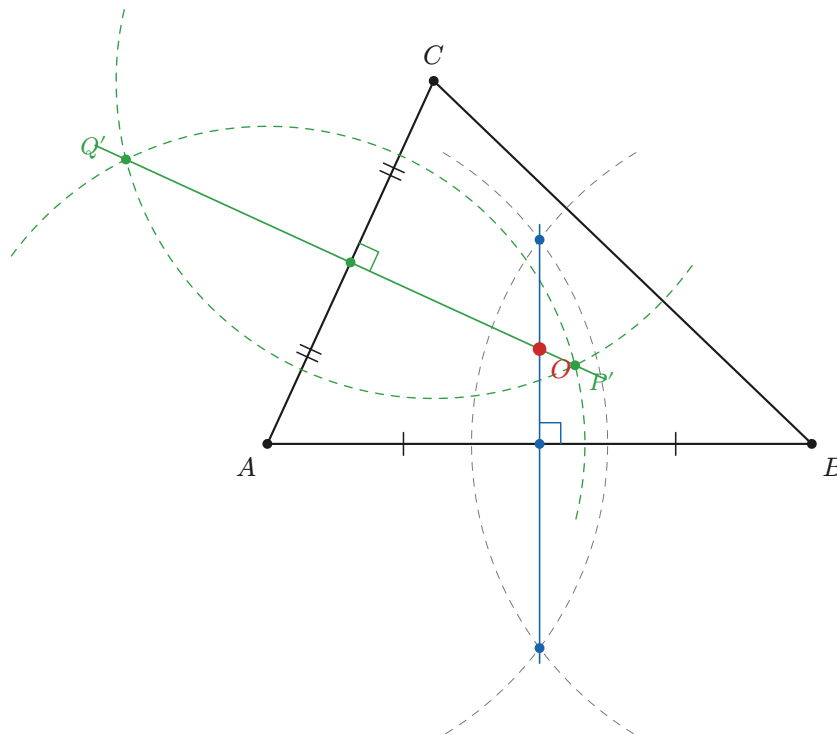
right-angle is not mini-square

mini-square is a **small set square**: it has a hypotenuse. Dropped into an angle, that hypotenuse cuts the corner with a diagonal. A right-angle mark is an **open square** — two sides, vertex held back from the corner. That is right-angle.

rotate on right-angle is the direction of the **first** side. Here (AB) is horizontal, so rotate: 0deg and the square rises into the triangle. On a side of angle θ , pass rotate: theta, then add 90° or turn it over depending on which side you want the square.

4.5 Step 5 — Perpendicular bisector of $[AC]$

Same gesture: **two arcs of equal radius** meeting at **two** points P' and Q' . Radius 4.2 cm exceeds $AC/2 \approx 2.64$. The intersection of the two perpendicular bisectors is O .

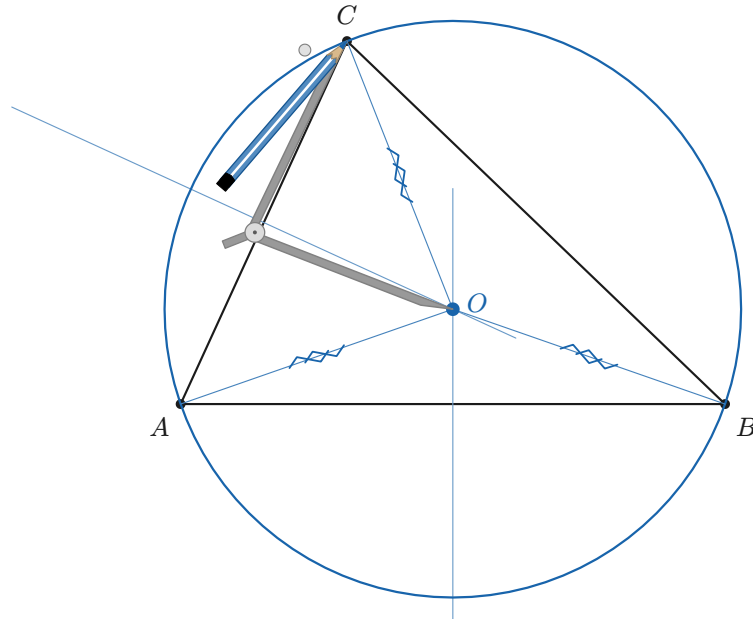


Each perpendicular bisector: **two** arcs, **two** intersection points. $(P'Q')$ meets (PQ) at O .

```
#let rAC = 4.2
#let (P2, Q2) = circ-inter(A, rAC, C, rAC)
#arc-through(A, P2, Q2) + arc-through(C, P2, Q2)
#let O = line-inter(P, Q, P2, Q2) // or circumcenter(A, B, C)
```

Mark $[AC]$ with **two** ticks, so it is not confused with $[AB]$ (one tick). The right angle at M_{AC} is turned inward by ra-in.

4.6 Step 6 — Compass at O , opening OA



The same two tildes on $[OA]$, $[OB]$ and $[OC]$: $OA = OB = OC$.

```
#let O = circumcenter(A, B, C)
#let R = dist(O, A)
#p-circle(O, R, stroke: rgb("#1864AB"), weight: 1.4)
#waves(O, A, n: 2) + waves(O, B, n: 2) + waves(O, C, n: 2)
#compass(O, C, scale: 0.52, pencil-colour: rgb("#1864AB"),
  pencil-lead: rgb("#1864AB"))
```

Without pencil-lead, the compass pencil would keep a **black** graphite tip — an ordinary pencil glued onto a blue barrel. On a coloured pencil, the lead matches the barrel.

Why O lies on all three perpendicular bisectors

By construction P and Q are equidistant from A and B , so every point of (PQ) is as well — in particular O . Likewise $OA = OC$. Hence $OA = OB = OC$: the circle centred at O through A passes through B and C .

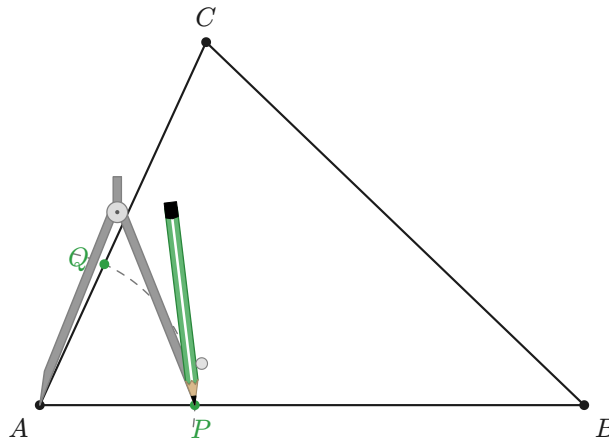
5 Incircle — with the compass

The **incircle** is tangent to the three sides. Its centre I is the intersection of the **angle bisectors**. An angle bisector is built with the compass in two moves: an arc centred at the vertex, then two equal arcs centred at the cut points.

What you mark

- **equal arcs** in the two half-angles (the bisector);
- at the points of tangency, a **right angle** between the radius and the side (the radius is perpendicular to the tangent);
- $IT_a = IT_b = IT_c$ (the inradius), carried by the circle itself.

5.1 Step 1 — Arc centred at A , cutting $[AB]$ and $[AC]$

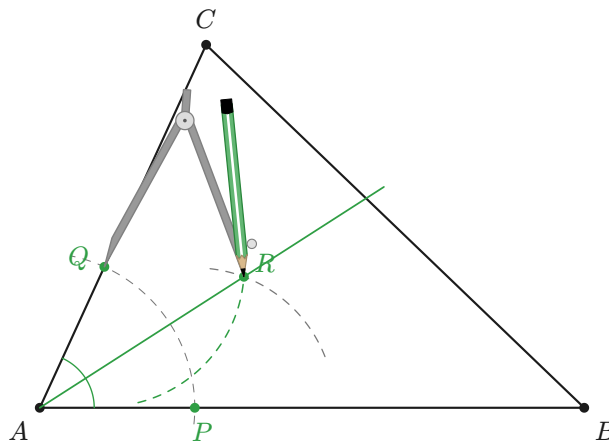


Needle at A . The arc cuts the two sides of the angle — **not** the third.

```
#let r = 2.05
#let P = lerp(A, B, r / dist(A, B)) // on [AB], distance r from A
#let Q = lerp(A, C, r / dist(A, C)) // on [AC], distance r from A
#p-arc(A, r, -8deg, 80deg, stroke: luma(120), dash: "dashed")
#compass(A, P, scale: 0.55)
```

5.2 Step 2 — Compass at P and at Q , same opening

The two arcs meet at R **inside** the angle, toward the right. The last arc is the one centred at Q : plant the needle at Q and stop the lead at R , not toward A .



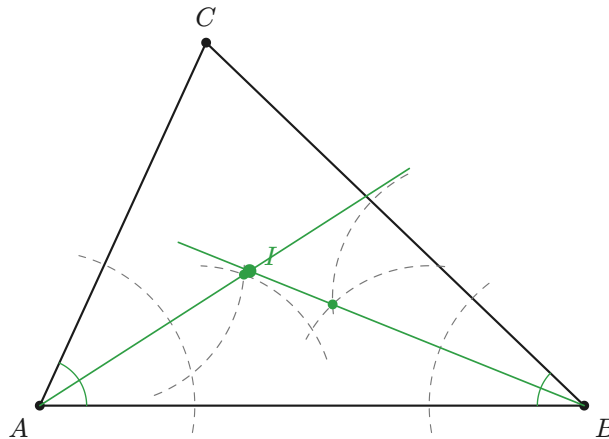
Needle at Q , lead at R : the pencil sits at the **end** of the arc, to the right.

```
#let R = farther(A, R1, R2) // the intersection farther from A
#arc-to(Q, R, back: 72deg, extra: 0deg)
#compass(Q, R, scale: 0.48) // lead at the end of the arc, to the right
```

Why it works: $AP = AQ$ (same arc), $PR = QR$ (same opening), AR common, so triangle $APR =$ triangle AQR (SSS) and the angles at A are equal.

5.3 Step 3 — Bisector of $\angle ABC$, then I

A second bisector is enough: the third goes through I . For **each** angle leave the **three** construction arcs: the arc at the vertex, then the two equal arcs that meet at R .

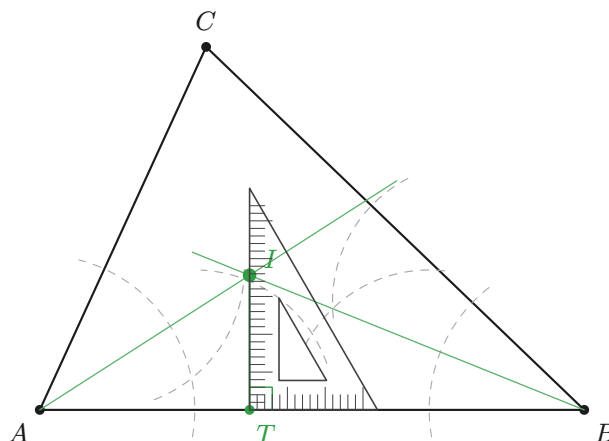


Three arcs at A , three arcs at B . I is their intersection.

```
#let I = line-inter(A, R-A, B, R-B) // or incenter(A, B, C)
```

5.4 Step 4 — Drop a perpendicular from I to a side, with the set square

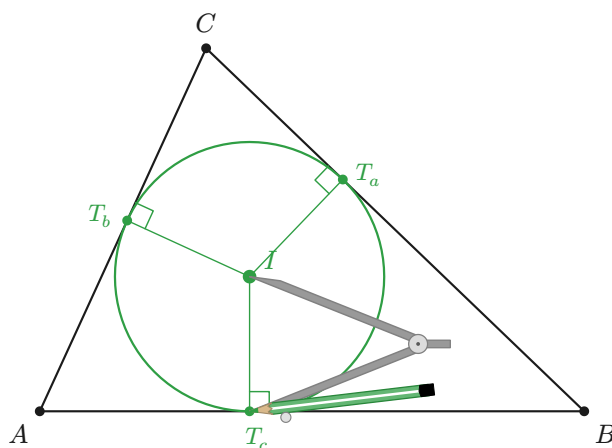
Keep the six bisector arcs. For the radius, do not draw an extra arc at C : drop the perpendicular from I to a side **with the set square**. The foot T is the point of tangency; IT will be the compass opening.



The 3+3 arcs stay. The set square on (AB) projects I to T : $(IT) \perp (AB)$.

```
#let T = foot(I, A, B)
#square-on(T, vsub(B, A), vsub(I, T), length: dist(I, T) + 1.15)
#ra-in(T, vsub(B, A), vsub(I, T), colour: rgb("#2F9E44"))
```

5.5 Step 5 — The circle, and the three tangencies



Three radii, three right angles: the circle is tangent to all three sides.

```
#let I = incenter(A, B, C)
#let T = foot(I, A, B)           // point of tangency on [AB]
#let r = dist(I, T)
#p-circle(I, r, stroke: rgb("#2F9E44"), weight: 1.4)
#ra-in(T, vsub(B, A), vsub(I, T), colour: rgb("#2F9E44"))
#compass(I, T, scale: 0.5, pencil-lead: rgb("#2F9E44"))
```

`ra-in(T, along, toward)` turns the square **inward**: `along` is the side, `toward` points to I . On $[AC]$, without that test, the square sat outside the triangle.

6 Centroid — medians, with the ruler

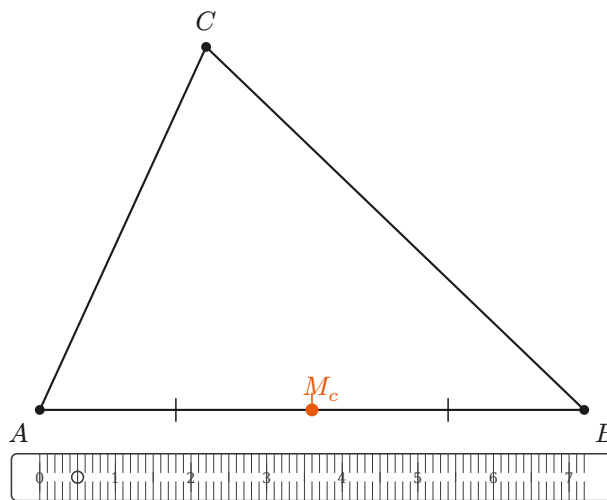
The **centroid** G (barycentre of the three vertices) is the intersection of the **medians**. A median joins a vertex to the **midpoint** of the opposite side. Find the midpoints with the ruler, then draw the three medians. The set square has no role here: a median is not a perpendicular.

What you mark

- on each side, **the same number of ticks** on both halves of the midpoint: one tick on $[AB]$, two on $[AC]$, three on $[BC]$;
- do **not** mark $AG = 2GM$ as a construction step (the $2:1$ split is a theorem). You may write it beside the figure once G is found.

6.1 Step 1 — Midpoint of $[AB]$ with the ruler

$AB = 7.2$ cm, the midpoint M_c is at 3.6 cm. Lay the ruler along the side, zero at A .



The 3.6 on the ruler falls on the midpoint. One tick on each half marks $AM_c = M_cB$.

```
#let Mc = midp(A, B)
#ruler(at: A, rotate: 0deg, length: dist(A, B),
      width: 0.62, clamp: false)
#ticks(A, Mc, n: 1) + ticks(Mc, B, n: 1)
```

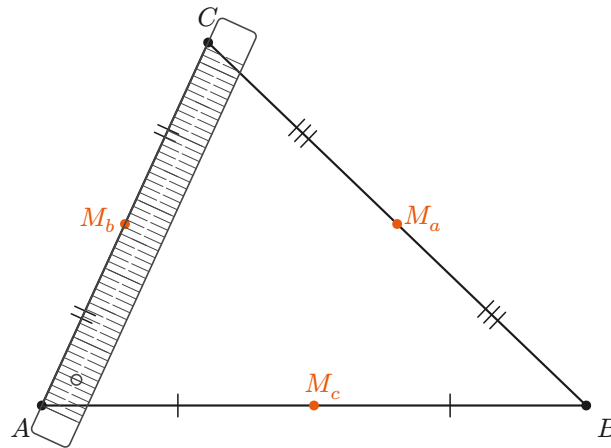
On an arbitrary side, **align** the ruler:

```
#ruler(
  at: A,
  rotate: vangle(vsub(C, A)),      // along [AC]
  length: dist(A, C),
  width: 0.62, clamp: false,
)
```

The ruler floor

width is floored at 1.5 cm by default, length at 3. A ruler laid **under** a 7 cm side, if it is 1.5 cm wide, covers half the triangle. `clamp: false` lifts both floors (minimum width then 0.05).

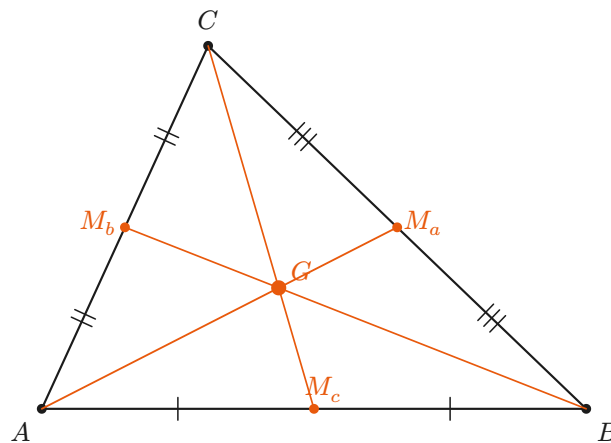
6.2 Step 2 — The three midpoints



Three midpoints, three distinct markings. The ruler lies on $[AC]$.

6.3 Step 3 — Draw the three medians

Join each vertex to the midpoint of the opposite side. Two medians suffice; the third is a check: if it misses G , a midpoint is wrong.



The three medians meet at G . No set square: this is not a right angle.

```
#p-line(A, Ma, stroke: rgb("#E8590C"), weight: 1.15, role: "edge")
#p-line(B, Mb, stroke: rgb("#E8590C"), weight: 1.15, role: "edge")
#p-line(C, Mc, stroke: rgb("#E8590C"), weight: 1.15, role: "edge")
```

Two medians are enough. The third is a check: if it misses G , a midpoint is wrong.

G divides each median in the ratio 2 : 1

$\overrightarrow{AG} = 2\overrightarrow{GM_a}$. In code, G is simply the average of the vertices — the same fact:

```
#let G = (
  (A.at(0) + B.at(0) + C.at(0)) / 3,
  (A.at(1) + B.at(1) + C.at(1)) / 3,
)
```

7 Orthocentre — altitudes, with the set square

The **orthocentre** H is the intersection of the **altitudes**. An altitude is the perpendicular from a vertex to the opposite side. That is **the** set-square gesture: one edge on the side, the other edge through the vertex.

What you mark

A **right-angle square** at each foot H_a , H_b , H_c . Nothing else: altitudes do not carry equal-length ticks.

7.1 Seating the set square on a side

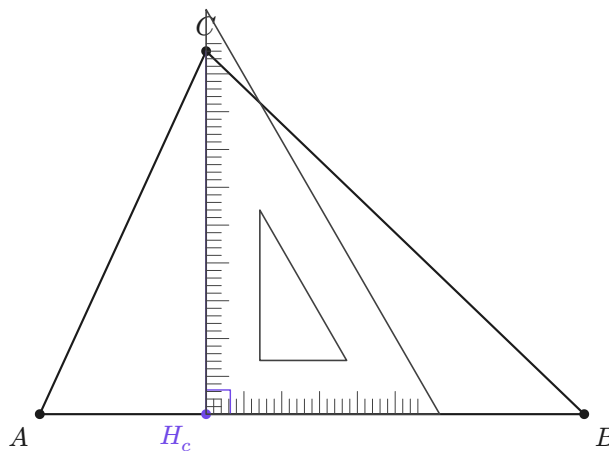
set-square has the right angle at the origin, the **base** (short leg) on $+x$, the **long leg** on $+y$. To seat it on a line:

1. **at**: the point where you want the right angle — in practice the **foot** of the altitude, or any point of the side while you still hunt for the foot;
2. **rotate**: the bearing of the side, $\text{vangle}(\text{vsub}(C, B))$ for (BC) ;
3. **flip**: if the long leg points away from the vertex, turn the square over.

The helper `square-on(at, along, toward)` does that test: `along` is a direction vector of the side, `toward` a vector toward the vertex. If local $+90^\circ$ is the wrong side, `flip` turns on.

7.2 Step 1 — Altitude from C , set square on (AB)

(AB) is horizontal, the perpendicular is vertical. The set square sits on AB , right angle at the foot $H_c = (2.2, 0)$. The long leg must **pass beyond** C : take length: $\text{dist}(C, H_c) + 0.55$.



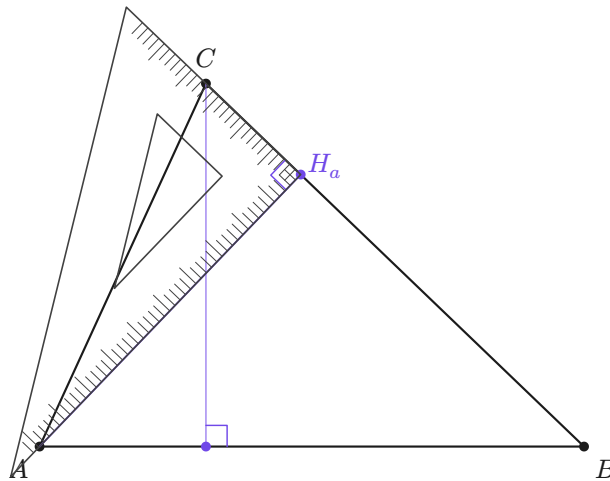
One edge on (AB) , the other through C . The foot H_c is read at the contact.

```
#let Hc = foot(C, A, B)
#square-on(Hc, vsub(B, A), vsub(C, Hc), length: dist(C, Hc) + 0.55)
#ra-in(Hc, vsub(B, A), vsub(C, Hc), colour: rgb("#7048E8"))
#p-line(C, Hc, stroke: rgb("#7048E8"), weight: 1.15, role: "edge")
```

On paper, the pupil **slides** the set square along (AB) until the other edge meets C , **then** draws. On the figure we seat it already in place — the foot is computed by `foot(C, A, B)`, the orthogonal projection.

7.3 Step 2 — Altitude from A , set square on (BC)

The side is no longer horizontal. `rotate` follows (BC) , `flip` tucks the square into the triangle.

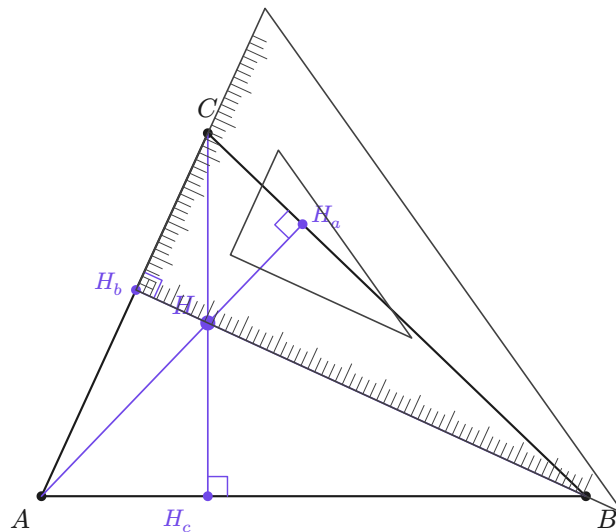


Set square on (BC) , long leg toward A . Two altitudes already give H .

```
#let Ha = foot(A, B, C)
#square-on(Ha, vsub(C, B), vsub(A, Ha), length: dist(A, Ha) + 0.55)
#ra-in(Ha, vsub(C, B), vsub(A, Ha), colour: rgb("#7048E8"))
```

If the small marking square **leaves** the triangle, rotate points into the wrong half-plane. Add 180° , or take $\text{vangle}(\text{vsub}(B, C))$ instead of $\text{vangle}(\text{vsub}(C, B))$.

7.4 Step 3 — The third altitude, and H



The three altitudes and their three right angles. H is interior: the triangle is acute.

```
#let H = line-inter(A, Ha, B, Hb) // or orthocenter(A, B, C)
```

Obtuse triangle

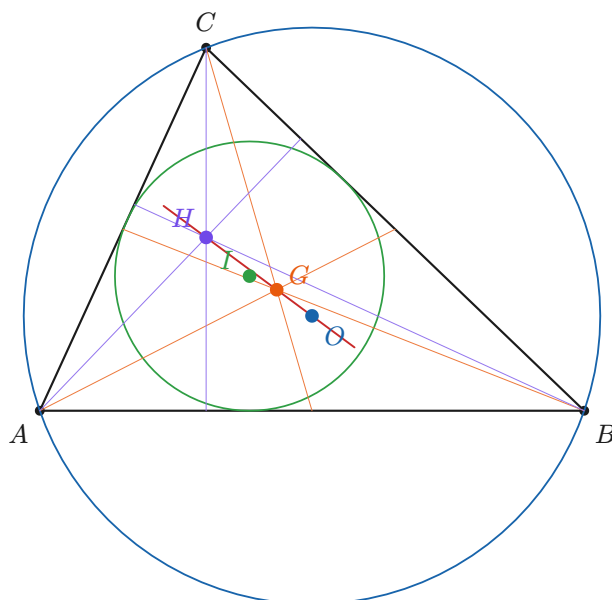
If an angle is obtuse, the orthocentre leaves the triangle and two feet fall on the **extensions**. `foot` still computes them (the projection parameter t is no longer in $[0, 1]$). Extend the side with dashed `line-ext`, and seat the set square on that extension.

8 The four centres together

On **every** triangle, O , G and H are collinear: that is **Euler's line**, and G lies one third of the way along $[OH]$ from O :

$$\overrightarrow{OG} = \frac{1}{3} \overrightarrow{OH}$$

I lies on that line only in isosceles triangles.



O , G , H collinear (red). I sits off the line: ABC is not isosceles.

8.1 Recap of the gestures

Centre	Lines	Instrument	Marking
O	perp. bisectors	compass, two equal arcs	$AM = MB$, right angle
I	angle bisectors	compass, arc then two arcs	angle arcs, radii $\perp$
G	medians	ruler (midpoints and medians)	equal-length ticks at midpoints
H	altitudes	set square on the side	right angle at each foot

9 Quick reference

9.1 geom

```
#geom(body, mode: "clean", roughness: 1.0, seed: 1,
      colour: black, frame: none, padding: 0.25)
```

body is an array of primitives. frame: (x0, x1, y0, y1) freezes the extent instead of fitting the content — useful for stacking steps **at the same scale**.

9.2 compass(from, to)

| Argument | Default | | — | — | — | | leg | 6.0 | length of a leg, cm | | scale | 1.0 | shrinks the instrument, **keeps** the feet | | flip | false | swings to the other side of [from to] | | pencil-colour | red | pencil barrel | | pencil-lead | auto (black) | lead; pass the same colour as the barrel | | show-pencil | true | |

If the span exceeds $2 \times \text{leg} \times \text{scale}$, the legs saturate and the feet no longer meet. Increase leg or bring the points closer.

9.3 set-square

| Argument | Default | | — | — | — | | at | (0,0) | **the right angle** | | rotate | 0deg | direction of the base (local +x) | | length | 10 | long leg; floor 4.5 if clamp: true | | flip | false | turn-over on the table | | values | true | numerals; false keeps the ticks | | clamp | true | |

square-on(at, along, toward) (this guide) chooses rotate and flip.

9.4 ruler

Zero at at, toward rotate. corner: 0.12 softens the ends (not a capsule). corner: 0: sharp corners. value-pos: "m" (middle), "h" (top), "b" (bottom, upside-down), combinable ("hb").

9.5 right-angle

```
#right-angle(at: vertex, rotate: 0deg, size: 0.32, colour: black)
```

Open square, vertex held back. rotate = direction of the first side.

9.6 Construction primitives

```
p-line(A, B, stroke: luma(120), weight: 1.0, dash: "dashed", role: "edge")
p-arc(centre, r, a0, a1, stroke: blue, dash: "dashed")
p-circle(centre, r, stroke: blue, weight: 1.3)
p-label(pos, [A], size: 10pt, fill: black)
```

role: "edge" (the tools' default) or "tick" / "detail": in rough mode, ticks wobble three times less, otherwise a 2 mm mark dissolves at the amplitude that clothes a 12 cm ruler.

9.7 Formulae used

- O is the intersection of the perpendicular bisectors.
- $I = \frac{aA+bB+cC}{a+b+c}$ with $a = BC$, $b = AC$, $c = AB$.
- $G = \frac{A+B+C}{3}$.
- H is the intersection of two altitudes; H_a is the projection of A onto side BC .

geomtools 0.1.0 — a port of Cédric Pierquet's OutilsGeomTikZ (LPPL 1.3c). He neither maintains nor endorses this port. Guide written for the triangle $A(0,0)$, $B(7.2,0)$, $C(2.2,4.8)$; the listings copy as-is once helpers.typ is imported.